

An almost square-root rate for the interior endpoint profile

Complete theorem; independently reviewed

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Theorem 1. *Let $p_1, \dots, p_m$ be nondecreasing functions on a finite ordered row set with positive probability weights w , and suppose*

$$\mathbb{E}_w \prod_{j \in S} p_j = \frac{1}{|S| + 1} \quad (S \subseteq [m]), \quad 0 \leq p_j \leq 1.$$

Put $u_j = 1 - p_j$ and $s = \sum_j u_j$. For every fixed $0 < \delta < L < \infty$, uniformly over columns and rows with $\delta \leq s \leq L$,

$$\max_j |mu_j - s| \leq C_{\delta, L} \sqrt{\frac{\log m}{m}}$$

for all sufficiently large m .

The constants are independent of the number of rows and of their positive weights. The fixed lower cutoff $\delta > 0$ is part of the statement; no estimate at the endpoint $s = 0$ is claimed here.

Inputs and their precise roles. This proof uses the independently reviewed notes

1. `../rational_designs/asymptotic/quantitative_uniform_endpoint_profile.tex`: on every fixed bounded s -interval, $\max_j |mu_j - s| \leq m^{-\alpha}$ for some fixed $\alpha > 0$, after decreasing the exponent to absorb logarithms;
2. `marked_cancellation_identity.tex`: the exact marked identity, centered coefficient estimate, all-tilt subset concentration, and localization-preserving subset comparisons;
3. `../localized_laguerre_kernel_derivatives.tex`: the all-orders localized complex-derivative bounds for the positive Laguerre Poisson kernel;
4. `../rational_designs/asymptotic/global_laguerre_spectral_truncation.tex`: global marked and unmarked residuals after truncation at differential order $J = O(\log m)$, and their off-annulus tail bounds;
5. `../size_bounds/localized_profile_maximum_principle.tex`: the localized mass-absorption lemma. We spell out the final weighted maximum argument below to handle the different deleted column coordinates explicitly.

No scalar transfer conjecture, rationality, or equal-weight assumption enters the argument.

Coordinates and probability measures. Fix a marked column i , and set

$$M = m - 1, \quad \ell = \lfloor M/4 \rfloor, \quad N = M - \ell, \quad z_j = \ell u_j.$$

For a uniformly selected ℓ -subset B of the other M indices, let R be its complement and put

$$A_B = \prod_{j \in B} p_j, \quad v_B = \frac{1}{N} \sum_{j \in R} z_j, \quad x_i = \frac{1}{M} \sum_{j \neq i} z_j, \quad h_i = z_i - x_i.$$

Write $\bar{A}_i = \mathbb{E}_B A_B$ and define the row measure

$$d\mu_i = (\ell + 1)\bar{A}_i dw.$$

It has total mass one by the mixed-moment identities. The scalar reference probability measure is

$$d\beta_\ell(x) = \frac{\ell + 1}{\ell}(1 - x/\ell)^\ell \mathbf{1}_{[0, \ell]}(x) dx.$$

Its density is bounded above and below on every fixed positive compact interval, uniformly for large m .

For a fixed large exponent A , use the choices in the global spectral-truncation theorem:

$$\tau = C_A \frac{\log m}{m}, \quad b = \sqrt{\tau}, \quad J = \lceil B_A \log m \rceil.$$

All constants are fixed before taking m large. Write $K_\tau(x, a)$ for the positive Laguerre Poisson kernel. On positive compact intervals it has Gaussian upper bounds at scale b and a lower bound c/b whenever $|x - a| \leq b$.

Lemma 1 (The two actual localized comparisons). *Fix a compact positive interval of test centers. There is a larger fixed compact interval $W \subset (0, \infty)$ of row coordinates such that, uniformly over marked columns and test centers, the following bounds hold. Let $H(q) = \max_j z_j(q) - \min_j z_j(q)$, including the marked column. Then*

$$|\mu_i(K_\tau(x_i, a)) - \beta_\ell(K_\tau(\cdot, a))| \leq \frac{C}{m\tau} \int_{x_i \in W} H^2 K_{c\tau}(x_i, a) d\mu_i + m^{-A}, \quad (1)$$

$$|\mu_i(h_i K_\tau(x_i, a))| \leq \frac{C}{m} \int_{x_i \in W} H^2(1 + b^{-1} + Hb^{-2}) K_{c\tau}(x_i, a) d\mu_i + m^{-A}. \quad (2)$$

Here c is a fixed enlargement factor. On W one has $H \leq R_m \leq m^{-\alpha}$ for some fixed $\alpha > 0$. Kernel tails outside W , with any fixed polynomially bounded prefactors used in these formulas, are negligible.

Proof. Choose W from the off-annulus theorem in the global truncation note, enlarging it if necessary. Its row coordinate there is $s_{-i} = \sum_{j \neq i} u_j = (M/\ell)x_i$; the conversion changes only fixed endpoints. On W , the finite balance bound $z_i \leq C(s_{-i} + 1)$ makes the full s bounded. The prior profile theorem therefore gives $H \leq m^{-\alpha}$ after weakening a power. All z_j then lie in a fixed positive compact interval, since their unmarked mean is x_i , and $p_j \geq 1/2$ for sufficiently large m .

The global unmarked differential residual compares the reference integral to

$$(\ell + 1)\mathbb{E}_w \mathbb{E}_B A_B \sum_{k=0}^J \frac{K_\tau^{(k)}(v_B, a)}{k!} a_k, \quad a_k = \frac{e_k((z_j - v_B)_{j \in R})}{\binom{N}{k}}.$$

The off-annulus theorem removes rows outside W with error m^{-A} , even with absolute values on each summand. On W , the centered local derivative estimate bounds the terms $k \geq 2$ by $CH^2/(m\tau)$ times a broader kernel. Average under the success-weighted subset law, then use its all-tilt concentration to keep that envelope at x_i . The localized subset Taylor comparison replaces $K_\tau(v_B, a)$ by $K_\tau(x_i, a)$ with the same bound. This proves (1).

For the marked identity the global residual is

$$(\ell + 1)\mathbb{E}_w \mathbb{E}_B A_B \left[(z_i - v_B) K_\tau(v_B, a) - \sum_{k=2}^J \left\{ \frac{(v_B - z_i) K_\tau^{(k)}(v_B, a)}{k!} + \frac{K_\tau^{(k-1)}(v_B, a)}{(k-1)!} \right\} a_k \right].$$

It is at most m^{-A} in absolute value. On W , the finite differential sum is bounded by $C[H^2/(mb) + H^3/(mb^2)]$ times a broader kernel, by the direct marked kernel lemma. The localized subset comparison replaces $(z_i - v_B)K_\tau(v_B, a)$ by $h_i K_\tau(x_i, a)$ with error $CH^2(1 + b^{-1} + Hb^{-2})/m$ times such a kernel. This proves (2).

The pure scalar kernels outside W have exponentially small Gaussian tails because their centers lie in the smaller fixed interval. Every row measure μ_i has mass one, and the row coordinates and the auxiliary factors are at most polynomial in m . Thus these scalar tails are negligible too. Increasing the initial exponent A absorbs the fixed number of residuals. $\square$

Uniform local mass. On W , $H \leq R_m$, so (1) is bounded by

$$\epsilon_m \mu_i(K_{c\tau}(x_i, a)) + m^{-A}, \quad \epsilon_m = \frac{CR_m^2}{m\tau} \longrightarrow 0.$$

Apply the mass-absorption lemma on an enlarged fixed annulus. Consequently, on any prescribed smaller annulus I , uniformly in i and the center a ,

$$\mu_i(|x_i - a| \leq b) \leq Cb, \quad c \leq \mu_i(K_\tau(x_i, a)) \leq C. \quad (3)$$

Broader kernel masses are also bounded. Gaussian integration over length- b intervals gives, for $r \geq b$ within that annulus,

$$\int_{|x_i - a| \geq r} K_\tau(x_i, a) d\mu_i \leq Ce^{-cr^2/b^2} + O(m^{-A}). \quad (4)$$

The outer annulus for this step is chosen strictly larger than I , so the constants remain uniform up to the boundaries of I .

A common coordinate for all marked columns. Put

$$y = \frac{\ell}{m}s, \quad h_i^* = z_i - y.$$

The exact identities

$$x_i = y - \frac{h_i^*}{M}, \quad h_i = \frac{m}{M}h_i^*, \quad H = \max_j h_j^* - \min_j h_j^* \quad (5)$$

will avoid comparing maxima in different deleted coordinates. Choose a fixed positive annulus $I = [a_-, a_+]$, and a strictly smaller I_0 , so that all target rows $\delta \leq s \leq L$ have $y \in I_0$ for large m . On a fixed enlargement of I the prior profile estimate gives $|h_i^*| \leq R_m \leq m^{-\alpha}$. Let $\phi(y) = \min(y - a_-, a_+ - y)$ on I , and set

$$D = \max_{i, \text{ rows with } y \in I} \phi(y)|h_i^*|.$$

If the row set here is empty, the target assertion is vacuous.

Suppose $D > C_*b$. Select a maximizing row and column i , write y_0 for its common coordinate, $\Delta = |h_i^*(y_0)|$, and $d_0 = \phi(y_0)$. Then

$$d_0 = D/\Delta \geq C_*b/R_m, \quad d_0/b \longrightarrow \infty$$

at a polynomial rate. Also $\Delta \geq c_I C_*b$. For every row with $|y - y_0| \leq d_0/2$, weighted maximality implies

$$|h_j^*(y)| \leq C\Delta \quad \text{for all } j, \quad H(y) \leq C\Delta. \quad (6)$$

Any row whose x_i lies in the fixed test annulus has bounded s_{-i} , and finite balance then makes its full s bounded. Thus it lies in a fixed y -enlargement where the prior profile estimate applies, even if it was not initially described by its y coordinate. Moreover, $|x_i - y| \leq R_m/M = o(b)$ on the entire enlarged annulus, by (5). Thus (6) holds on a fixed smaller x_i -neighborhood of the selected deleted coordinate $x_0 = y_0 - h_i^*(y_0)/M$. The cutoff weights there are comparable to d_0 . Rows not in this neighborhood are separated from x_0 by a fixed multiple of d_0 , apart from negligible $o(b)$ shifts.

A one-sided positive test. Choose fixed B_0 large enough for the Gaussian tails in (4), and then set $A_0 = 2B_0$. If the selected deviation is positive, center the kernel at $a = x_0 + A_0b$; if it is negative, take $a = x_0 - A_0b$. Both centers remain inside I . Consider the positive case. Whenever $|x_i - a| \leq B_0b$, one has $x_i > x_0$ strictly. The monotone ordering of the columns then gives $z_i \geq z_i(x_0)$, even if there are other flat rows. Therefore

$$h_i(x_i) \geq h_i(x_0) - (x_i - x_0) \geq \frac{m}{M}\Delta - (A_0 + B_0)b \geq \Delta/2,$$

provided C_* was chosen large enough. For a negative selected deviation the same argument to the left gives $h_i \leq -\Delta/2$.

The kernel mass on this favorable window is bounded below by a positive constant using (3)–(4), after choosing B_0 sufficiently large. In the surrounding $d_0/4$ neighborhood, (6) bounds $|h_i|$ by $C\Delta$, so its opposite-sign contribution outside the favorable window is as small a fixed multiple of Δ as desired. Outside that neighborhood the Gaussian factor is $\exp[-c(d_0/b)^2]$. Since d_0/b grows polynomially, every polynomial row factor is absorbed. It follows that

$$|\mu_i(h_i K_\tau(x_i, a))| \geq c\Delta. \tag{7}$$

On the other hand, (2), the local range bound, bounded broader kernel mass, and the same negligible far tails give

$$|\mu_i(h_i K_\tau(x_i, a))| \leq \frac{C\Delta^2}{m}(1 + b^{-1} + \Delta b^{-2}) + m^{-A} = o(\Delta).$$

Indeed $mb^2 = C_A \log m$, $mb \rightarrow \infty$, and $\Delta \leq R_m \rightarrow 0$, whereas $\Delta \geq c_I C_* b$. This contradicts (7). Hence $D \leq C_* b$. On I_0 the cutoff is bounded below, so $|h_i^*| \leq Cb$. Finally $mu_i - s = (m/\ell)h_i^*$ and m/ℓ is bounded. This proves Theorem 1.